Annex I. Public Debt Sustainability

Bottomline: Paraguay's debt is sustainable, and risks to a sustainable path are low.

Baseline

Baseline public debt stood at 26 percent of GDP as of 2019. It will jump to about 32 percent by 2021 and then gradually converge toward 30 percent in the medium term.	The main driver of debt is the fiscal response to the Covid-19 crisis in 2020, followed by a gradual consolidation over the next three years.

Stress Tests

Stress tests show that the most adverse shock would be a real GDP shock. Even under this shock, debt would remain safely below 50 percent, peaking at 36.8 percent of GDP in 2022. A one-time real-exchange rate shock of similar magnitudes as of 2015, when commodity prices decreased sharply, would have debt peak at 36.1 percent in 2021.	Real GDP growth in Paraguay is volatile, due to the impact of shifting weather conditions on agriculture. Agricultural commodity price fluctuations are the most important source of real-exchange rate shocks in Paraguay. While the share of foreign-currency denominated debt is high, short-term capital flows are low, and the Central Bank's reserves are high. Interest-rate risk is limited, as Paraguay's external sovereign debt consists of long-term fixed-coupon bonds, the first one of which needs to be refinanced in 2023.

Assumptions

The baseline assumes negative growth (-1.0 percent) in 2020, followed by a rebound to 4 percent in 2021. In the medium-term growth converges to potential output growth of 3.5 to 4.0 percent. The fiscal deficit is projected to leap to 4.5 percent of GDP in 2020 and then gradually converge back to 1.5 percent by 2023.	The growth assumption for 2020 is very conservative, as a large share of agriculture output has been brought in already in the first quarter, leading to a sharp rebound of this sector from last year's drought. The projected contraction of services and manufacturing underlying the growth forecast is very sharp. The unwinding of this shock is expected to produce a strong rebound in 2021.

Coverage and contingent liabilities

Debt coverage comprises the consolidated public sector, and all domestic key entities and liabilities are included. Public sector debt does not include liabilities by the binational entities Itaipú and Yacyretá.	There are no known contingent liabilities.

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Paraguay: Public Sector Debt Sustainability Analysis - Baseline Scenario

(in percent of GDP unless otherwise indicated)

Debt, Economic and Market Indicators

	Actual			Projections					
	2009-2017	2018	2019	2020	2021	2022	2023	2024	2025
Nominal gross public debt	15.0	22.2	26.3	30.6	31.8	31.3	30.6	30.3	30.2
Public gross financing needs	1.4	2.5	4.7	5.7	4.1	3.3	4.2	2.6	2.7
Real GDP growth (in percent)	4.5	3.7	0.2	-1.0	4.0	4.0	4.0	4.0	3.5
Inflation (GDP deflator, in percent)	3.7	2.1	-0.3	8.1	1.9	3.5	2.9	2.5	2.6
Nominal GDP growth (in percent)	8.3	5.9	-0.1	7.0	6.0	7.6	7.0	6.6	6.2
Effective interest rate (in percent)	4.9	5.8	5.8	2.6	2.5	2.8	2.9	3.3	3.6

Contribution to Changes in Public Debt

	Actual			Projections					
	2009-2017	2018	2019	2020	2021	2022	2023	2024	2025
Change in gross public sector debt	0.6	2.3	4.1	4.3	1.2	-0.5	-0.6	-0.3	-0.1
Identified debt-creating flows	0.7	3.3	5.6	4.0	1.0	-0.6	-0.7	-0.5	-0.4
Primary deficit	-0.2	0.6	2.6	3.9	2.4	1.2	0.6	0.6	0.6
Primary (noninterest) revenue and grants	17.3	18.8	19.6	19.2	19.2	19.6	20.0	20.0	20.0
Primary (noninterest) expenditure	17.1	19.4	22.2	23.1	21.6	20.8	20.6	20.6	20.6
Automatic debt dynamics	-0.3	1.0	2.9	-1.1	-1.0	-1.4	-1.2	-0.9	-0.8
Interest rate/growth differential	-0.4	0.0	1.3	-1.1	-1.0	-1.4	-1.2	-0.9	-0.8
Of which: real interest rate	0.2	0.7	1.3	-1.3	0.2	-0.2	0.0	0.2	0.2
Of which: real GDP growth	-0.6	-0.7	0.0	0.2	-1.2	-1.2	-1.2	-1.1	-1.0
Exchange rate depreciation	0.2	1.0	1.6	...	...	...	...	...	...
Other identified debt-creating flows	1.1	1.7	0.1	1.2	-0.4	-0.4	-0.2	-0.2	-0.2
NFPS asset accumulation	1.1	1.7	0.1	1.2	-0.4	-0.4	-0.2	-0.2	-0.2
Residual	-0.1	-0.9	-1.4	0.4	0.2	0.1	0.1	0.2	0.3

Paraguay Public DSA Risk Assessment

Heat Map

Debt level	Real GDP Growth Shock	Primary Balance Shock	Real Interest Rate Shock	Exchange Rate Shock	Contingent Liability shock
Gross financing needs	Real GDP Growth Shock	Primary Balance Shock	Real Interest Rate Shock	Exchange Rate Shock	Contingent Liability Shock
Debt profile	Market Perception	External Financing Requirements	Change in the Share of Short-Term Debt	Public Debt Held by Non-Residents	Foreign Currency Debt

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Figure 1. Paraguay: Public Sector Debt Sustainability Analysis
(In percent of GDP unless otherwise specified)

Debt, Economic and Market Indicators [1/]

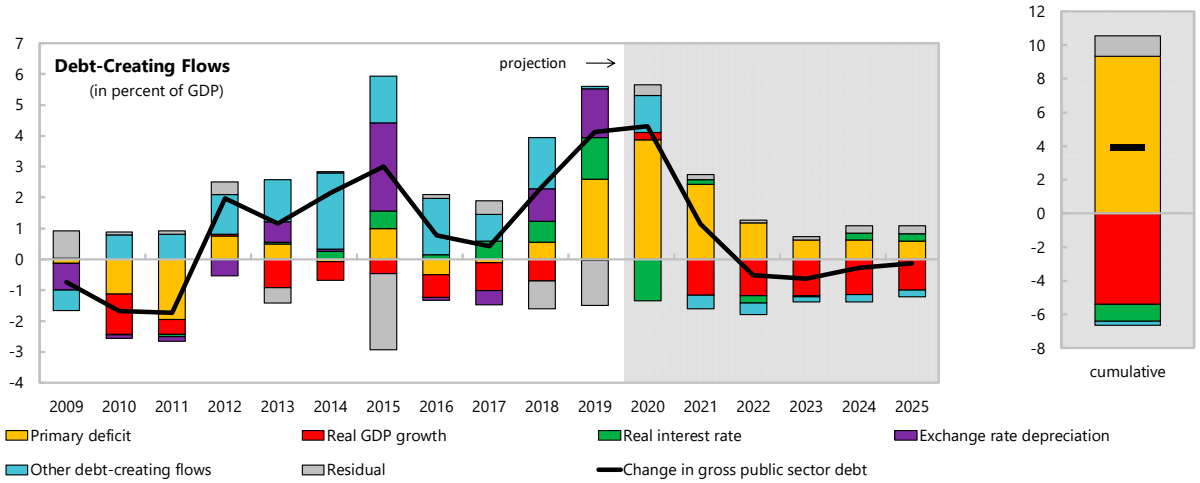
	Actual			Projections					
	2009-2017	2018	2019	2020	2021	2022	2023	2024	2025
Nominal gross public debt	15.0	22.2	26.3	30.6	31.8	31.3	30.6	30.3	30.2
Public gross financing needs	1.4	2.5	4.7	5.7	4.1	3.3	4.2	2.6	2.7
Real GDP growth (in percent)	4.5	3.7	0.2	-1.0	4.0	4.0	4.0	4.0	3.5
Inflation (GDP deflator, in percent)	3.7	2.1	-0.3	8.1	1.9	3.5	2.9	2.5	2.6
Nominal GDP growth (in percent)	8.3	5.9	-0.1	7.0	6.0	7.6	7.0	6.6	6.2
Effective interest rate (in percent) [2/]	4.9	5.8	5.8	2.6	2.5	2.8	2.9	3.3	3.6

As of March 19, 2020

Sovereign Spreads		
EMBIG (bp)		372
5Y CDS (bp)		...
Ratings	Foreign	Local
Moody's	Ba1	Ba1
S&Ps	BB	BB
Fitch	BB+	BB+

Contribution to Changes in Public Debt

	Actual			Projections							
	2009-2017	2018	2019	2020	2021	2022	2023	2024	2025	cumulative	debt-stabilizing primary balance [8/]
Change in gross public sector debt	0.6	2.3	4.1	4.3	1.2	-0.5	-0.6	-0.3	-0.1	3.9	
Identified debt-creating flows	0.7	3.3	5.6	4.0	1.0	-0.6	-0.7	-0.5	-0.4	2.7	
Primary deficit	-0.2	0.6	2.6	3.9	2.4	1.2	0.6	0.6	0.6	9.3	-1.0
Primary (noninterest) revenue and grants	17.3	18.8	19.6	19.2	19.2	19.6	20.0	20.0	20.0	118.0	
Primary (noninterest) expenditure	17.1	19.4	22.2	23.1	21.6	20.8	20.6	20.6	20.6	127.4	
Automatic debt dynamics [3/]	-0.3	1.0	2.9	-1.1	-1.0	-1.4	-1.2	-0.9	-0.8	-6.4	
Interest rate/growth differential [4/]	-0.4	0.0	1.3	-1.1	-1.0	-1.4	-1.2	-0.9	-0.8	-6.4	
Of which: real interest rate	0.2	0.7	1.3	-1.3	0.2	-0.2	0.0	0.2	0.2	-1.0	
Of which: real GDP growth	-0.6	-0.7	0.0	0.2	-1.2	-1.2	-1.2	-1.1	-1.0	-5.4	
Exchange rate depreciation [5/]	0.2	1.0	1.6	...	...	...	...	...	...	...	
Other identified debt-creating flows	1.1	1.7	0.1	1.2	-0.4	-0.4	-0.2	-0.2	-0.2	-0.2	
NFPS asset accumulation [6/]	1.1	1.7	0.1	1.2	-0.4	-0.4	-0.2	-0.2	-0.2	-0.2	
Residual [7/]	-0.1	-0.9	-1.4	0.4	0.2	0.1	0.1	0.2	0.3	1.2	



Source: Fund staff estimates and projections.

1/ Public sector is defined as consolidated public sector. It includes the non-financial public sector and the central bank. The stock of public debt excludes central bank bills.

2/ Defined as interest payments divided by debt stock (excluding guarantees) at the end of previous year.

3/ Derived as [(r - π(1+g) - g + ae(1+r))/(1+g+π+gπ)) times previous period debt ratio, with r = interest rate; π = growth rate of GDP deflator; g = real GDP growth rate; a = share of foreign-currency denominated debt; and e = nominal exchange rate depreciation (measured by increase in local currency value of U.S. dollar).

4/ The real interest rate contribution is derived from the numerator in footnote 3 as r - π (1+g) and the real growth contribution as -g.

5/ The exchange rate contribution is derived from the numerator in footnote 3 as ae(1+r).

6/ Includes social security surplus, accumulation of deposits from the sovereign bond issuance in 2014, and financing of the national development bank.

7/ Includes asset changes and interest revenues (if any). For projections, it includes the impacts of exchange rate changes.

8/ Assumes that key variables (real GDP growth, real interest rate, and other identified debt-creating flows) remain at the level of the last projection year.

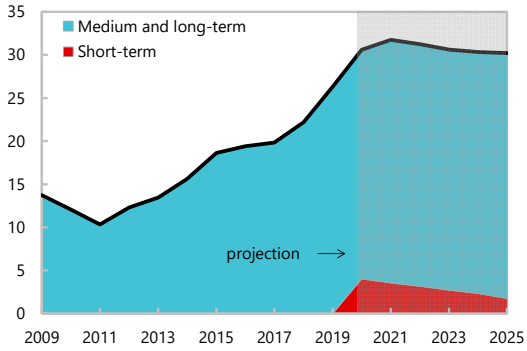
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Paraguay: Public DSA - Composition of Public Debt and Alternative Scenarios

Composition of Public Debt

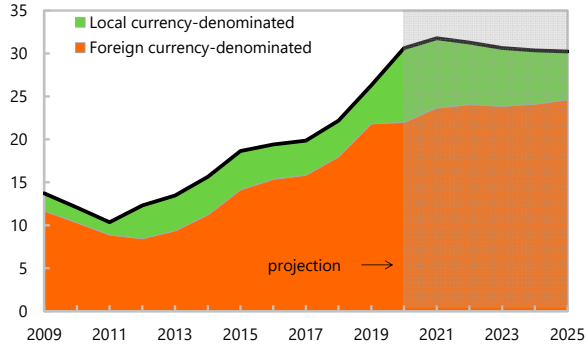
By Maturity

(in percent of GDP)

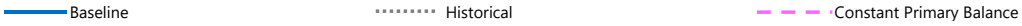


By Currency

(in percent of GDP)

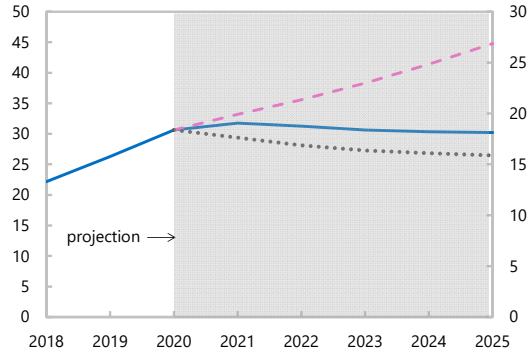


Alternative Scenarios



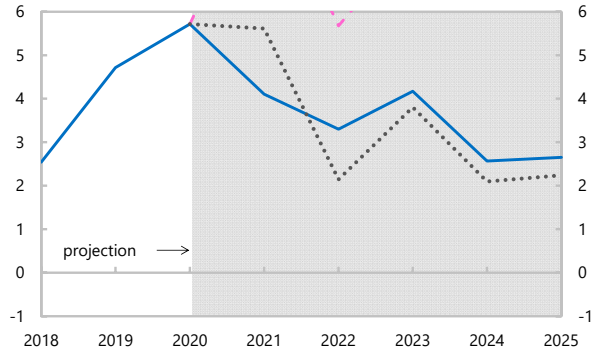
Gross Nominal Public Debt

(in percent of GDP)



Public Gross Financing Needs

(in percent of GDP)



Underlying Assumptions

(in percent)

Baseline Scenario	2020	2021	2022	2023	2024	2025
Real GDP growth	-1.0	4.0	4.0	4.0	4.0	3.5
Inflation	8.1	1.9	3.5	2.9	2.5	2.6
Primary Balance	-3.9	-2.4	-1.2	-0.6	-0.6	-0.6
Effective interest rate	2.6	2.5	2.8	2.9	3.3	3.6
Constant Primary Balance Scenario						
Real GDP growth	-1.0	4.0	4.0	4.0	4.0	3.5
Inflation	8.1	1.9	3.5	2.9	2.5	2.6
Primary Balance	-3.9	-3.9	-3.9	-3.9	-3.9	-3.9
Effective interest rate	2.6	2.5	3.5	3.6	4.1	4.4

Historical Scenario	2020	2021	2022	2023	2024	2025
Real GDP growth	-1.0	4.4	4.4	4.4	4.4	4.4
Inflation	8.1	1.9	3.5	2.9	2.5	2.6
Primary Balance	-3.9	-0.2	-0.2	-0.2	-0.2	-0.2
Effective interest rate	2.6	2.5	3.7	3.9	4.6	4.9

Source: Fund staff estimates and projections.

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